DR. NURDIANA ZAINOL ABIDIN

SENIOR LECTURER (DS13) | EXERCISE PHYSIOLOGY | HEALTHY AGEING | DIGITAL HEALTH

Department of Community Health, Pusat Kanser Tun Abdullah Ahmad Badawi (PKTAAB), Universiti Sains Malaysia | Bertam, Penang, Malaysia

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Scopus Author ID 56366272800

Researcher ID O-3964-2016

Web of Science ResearcherID HTM-3927-2023

Google Scholar ID C_jBuOYAAAAJ

ACADEMIC PROFILE

Research scientist and senior lecturer specialising in musculoskeletal ageing, sarcopenia, osteosarcopenic obesity, clinical exercise physiology and technology-enabled rehabilitation. Leads research in immersive virtual reality for mild cognitive impairment, dissolving microneedle delivery systems, and epidemiological and machine-learning approaches to musculoskeletal screening. Experience spans competitive research grants, postgraduate supervision, curriculum leadership, educational innovation and public science communication.

EMPLOYMENT HISTORY AND ACADEMIC APPOINTMENTS

Apr 2021 - present — **Senior Lecturer (DS13) | Universiti Sains Malaysia**
Department of Community Health, Pusat Kanser Tun Abdullah Ahmad Badawi (PKTAAB; formerly Advanced Medical and Dental Institute), Bertam, Penang. Teaching, research, supervision, grants and academic service.

May - Oct 2015 — **Research Associate (Contract) | Universiti Sains Malaysia**
Lifestyle Science Cluster, Advanced Medical and Dental Institute (AMDI), Penang. Research activities in exercise, health and active ageing.

Oct 2009 - Oct 2012 — **Research Officer | Universiti Sains Malaysia**
Lifestyle Science Cluster, Advanced Medical and Dental Institute (AMDI), Penang. Research coordination and execution of community, physical activity and physiology studies.

Nov 2008 - May 2009 — **Product Technologist (Management Trainee) | Nestlé Manufacturing Malaysia**
Shah Alam, Selangor. Research and development of ready-to-drink products; production mixing/filling trials, product testing, material coordination and technical-release processes.

Concurrent leadership and research affiliation

Head of Programme | MSc Clinical Exercise Science (Coursework), USM (2023-2026). Led academic sessions 2023/2024, 2024/2025 and 2025/2026; appointment ended September 2026.

Dementia Multidisciplinary Research Programme | PKTAAB, USM. Research group member and Principal Investigator for immersive VR-based dual-task training in older adults with mild cognitive impairment.

ACADEMIC QUALIFICATIONS

2021 — **PhD in Biosciences | University of Nottingham (Malaysia / UK)**
October 2016 - February 2021. Thesis: Osteosarcopenic Obesity: The Development of Screening Test Criteria and the Association with Bioavailable 25(OH)D. Supervisor: Assoc. Prof. Soma Mitra.

2015 — **MSc in Health Sciences (by Research) | Universiti Sains Malaysia**
November 2012 - April 2015. Thesis: Physical Activity Levels of Malay Older Adults Objectively Measured Using Tri-axial Accelerometer and Its Associated Factors. Supervisor: Prof. Rabindarjeet Singh; international advisor: Prof. Wendy Brown (University of Queensland).

2008 — **BSc (Hons) Biomedical Sciences | International Islamic University Malaysia (IIUM)**
September 2004 - June 2008. Final-year project on the antibacterial activity of ten medicinal plants from Kedah and Penang. Supervisor: Assoc. Prof. Othman Abd Samah.

RESEARCH EXPERTISE

- Musculoskeletal ageing; sarcopenia and osteosarcopenic obesity; bone health; body composition and vitamin D
- Clinical exercise physiology; exercise immunology; functional assessments and rehabilitation
- Immersive virtual reality; motor-cognitive dual-task training; dementia and mild cognitive impairment
- NHANES; clinical prediction modelling; machine learning; epidemiology; systematic review and meta-analysis
- Dissolving microneedle formulation; resveratrol and poorly soluble bioactive compounds
- Digital teaching innovation, biomechanics and educational technology

RESEARCH FUNDING AND GRANTS

2022 | RM171,800 — **Fundamental Research Grant Scheme (FRGS), Ministry of Higher Education Malaysia**
Principal Investigator. Mechanistic evaluation of resveratrol-loaded dissolvable microneedle patches and their potential relevance to delayed onset muscle soreness following exercise.

2024 | RM45,070 — **Universiti Sains Malaysia Short-Term Grant**
Competitive institutional short-term research funding awarded in 2024.

2012-2014 | RM40,000 **AMDI Postgraduate Research Student Fund, USM**
Postgraduate research funding.

2012-2014 | RM10,000 **Postgraduate Research Scheme, Institute of Postgraduate Studies, USM**
Postgraduate research support.

PRINCIPAL RESEARCH PROJECTS AND COLLABORATIONS

Resveratrol dissolving microneedle delivery for exercise-related muscle recovery (2022-2026). Principal Investigator. PVP-VA/PEG 400 dissolving microneedles fabrication; formulation, characterisation and preliminary permeation investigation. Institutional invention disclosure reviewed in 2026. Patent protection is under evaluation, not yet a granted patent.

Immersive VR motor-cognitive training in older adults with mild cognitive impairment (2022-2026). Principal Investigator, Dementia Multidisciplinary Research Programme. Fully immersive seated VR intervention compared with cycling; cognitive performance, physical function and functional MRI outcomes.

Clinical prediction and population-based musculoskeletal research (2026). NHANES research on screening for low bone mineral density in postmenopausal women with obesity and low appendicular lean mass across body sizes; survey-aware statistical and machine-learning methods. Manuscripts in development.

Additional research and methods development. Research concepts and proposals in grip-function testing, AI-assisted sarcopenia risk assessment, immunometabolic muscle-bone signalling and transdermal delivery of plant-derived lipophilic compounds. These are in the development stage.

Earlier multi-investigator research collaborations (2010-2013). Contributed to community healthy-lifestyle intervention research; a KAO Corporation (Japan) collaboration on sanitary products and stress responses; football-player hydration studies; and caloric restriction / fasting and cognitive function research.

PEER-REVIEWED JOURNAL PUBLICATIONS

20 full-length journal articles and reviews (2009-2026), presented in reverse chronological order. Conference abstracts are listed separately.

01 Abidin NZ, Mohd Shariff N, Zamri EN, Shamsuddin S. Beyond predictive performance: A systematic review and critical methodological appraisal of AI/ML and conventional modelling strategies in breast, colorectal, and pancreatic cancer. Artificial Intelligence in Medicine. 2026;181:103506. https://doi.org/10.1016/j.artmed.2026.103506

02 Wan Ahmad WNI, Mohd Razali ZN, Baharum MA, Abidin NZ, Ismail SA, Kamat SR, Shamsuddin S. Limb guidance system in robot-assisted gait training for clinical functional improvement in stroke patients: A narrative review. Pan African Medical Journal. 2026;54(Suppl 1):18. https://doi.org/10.11604/pamj.supp.2026.54.1.51612

03 Abidin NZ, Mansor NS, Yusof HA, Abd Hamid AI, Mat Isa S. The impact of dual-task training versus single-task training on cognitive and physical function in neurologically impaired individuals: A systematic review and meta-analysis of randomized controlled trials. Perceptual and Motor Skills. 2026;Online first, 12 February 2026. https://doi.org/10.1177/00315125261425126

04 Huang W, Abidin NZ, Mansor NS, Yusof HA, Lian B. Beyond conventional mental skills training in sports: A critical review of current paradigms. Malaysian Journal of Sport Science and Recreation. 2026;22(1):31-76. https://doi.org/10.24191/mjssr.v22i1.8233

05 Wen F, Yusof HA, Abidin NZ. Effects of active video games on affective responses and mental health symptoms in young adults: A systematic review. BMC Public Health. 2025;25:3995. https://doi.org/10.1186/s12889-025-25413-1

06 Wen F, Yusof HA, Abidin NZ. Effects of active video games on energy expenditure and weight management among young adults: A systematic review. Games for Health Journal. 2025;14(5):345-357. https://doi.org/10.1089/g4h.2025.0044

07 Li MZ, Zamri E, Liu K, Abidin NZ, Mei X. Fall occurrence and association with knowledge and practice on fall prevention among elderly residents in nursing homes, PR China. Southeast Asian Journal of Tropical Medicine and Public Health. 2025;56(4):489-517.

08 Abidin NZ, Ooi CH, Nosaka K, et al. Effects of resveratrol supplementation on delayed onset muscle soreness and muscle recovery: A systematic review. Malaysian Journal of Medical Sciences. 2024;31(6):77-102. https://doi.org/10.21315/mjms2024.31.6.7

09 Abidin NZ. The "fat or fit paradox" in postmenopausal women: A cross-sectional study on the association between skeletal muscle mass, adiposity, and muscle strength. Muscles, Ligaments and Tendons Journal. 2023;13(3):494-503. https://doi.org/10.32098/mltj.03.2023.20

10 Abidin NZ. Comparison of muscle, bone and fat indices between stages of sarcopenia in postmenopausal Malaysian women. Malaysian Journal of Medical Sciences. 2023;30(5):91-105. https://doi.org/10.21315/mjms2023.30.5.8

11 Abidin NZ. Obesity and musculoskeletal health of young and older Malaysian women: A cross-sectional study. Malaysian Journal of Medical Sciences. 2023;30(1):137-151. https://doi.org/10.21315/mjms2023.30.1.12

12 Mitra SR, Abidin NZ. Comparison of anthropometric profile, bone-related parameters, biochemical parameters, functional capacity and vitamin D status: A study on postmenopausal Malaysian women. Malaysian Journal of Nutrition. 2021; https://doi.org/10.31246/mjn-2020-0080

13 Abidin NZ, Mitra SR. Determination of cut-off values for the screening of osteosarcopenia in obese postmenopausal women. Current Gerontology and Geriatrics Research. 2021;2021:6634474. https://doi.org/10.1155/2021/6634474

14 Abidin NZ, Mitra SR. Total vs. bioavailable: Determining a better 25(OH)D index in association with bone density and muscle mass in postmenopausal women. Metabolites. 2021;11(1):23. https://doi.org/10.3390/metabo11010023

15 Abidin NZ, Brown W, Clark B, Muhamed AMC, Singh R. Physical activity measurement by accelerometry among older Malay adults living in semi-rural areas: Feasibility study. Journal of Aging and Physical Activity. 2016;24(4):533-539.

16 Abidin NZ, Brown W, Clark B, Muhamed AMC, Singh R. Assessing the physical activity level of elderly Malays living in semi-rural areas using the tri-axial accelerometer. Malaysian Journal of Nutrition. 2016;22(3):363-374.

17 Abidin NZ, Muhamed AMC, Singh R. Associations between tri-axial accelerometer-determined physical activity and self-reported health-related quality of life of older Malay adults using cut-

points for vector magnitude. Malaysian Journal of Movement, Health & Exercise. 2016;5(1). https://doi.org/10.15282/mohe.v5i1.102

18 Abidin NZ, Muhamed AMC, Singh R. Physical activity level and sedentary behavior of Malay adults objectively measured using a tri-axial accelerometer: A preliminary study. Asian Journal of Exercise & Sports Science. 2014;11(2):56-66.

19 Mohamed NG, Abidin NZ, Law KS, Abe M, Suzuki M, Muhamed AMC, Singh R. The effect of wearing sanitary napkins of different thicknesses on physiological and psychological responses in Muslim females. Journal of Physiological Anthropology. 2014;33:28. https://doi.org/10.1186/1880-6805-33-28

20 Samah OA, Abidin NZ. Antibacterial activity of ten medicinal plants obtained from some selected villages in the states of Kedah and Penang, Malaysia. Indonesian Journal of Pharmacy. 2009;20(2):99-103.

BOOK CHAPTER AND PUBLISHED CONFERENCE ABSTRACT

Book chapter | 2023 Abidin NZ. Osteosarcopenic Obesity: A Comprehensive Guide to Prevention and Management. In: Shaharudin S, Atan Edinur HA, Che Mat NF (eds.), Genetic Disorders and Rare Diseases: Current Updates. New York: Nova Science Publishers; 2023. Chapter 1, pp. 1-28. https://doi.org/10.52305/WNGG1520

Published conference abstract | 2022 Abidin NZ, Mitra SR. Osteosarcopenic obesity: The interrelationship between fat, bone and muscle and their associations with total and bioavailable 25(OH)D levels in postmenopausal women. Journal of Frailty & Aging, 11 (conference abstract). https://doi.org/10.14283/jfa.2022.23

POSTGRADUATE RESEARCH SUPERVISION

Research supervision: 2 main-supervisor appointments (1 PhD; 1 MSc by Research) and 4 co-supervisor appointments (2 PhD; 2 MSc by Research). Li Muzheng completed the MSc in 2025; Tie Yuhe passed the MSc viva in October 2026, with minor corrections.

Huang Weijie | Main supervisor | PhD (2022-present)
Psychological skills training, mental toughness, salivary cortisol, physiological load and simple reaction time among collegiate basketball athletes.

Tie Yuhe | Main supervisor | MSc Research (2024-2026)
Tai Chi intervention, Type A behaviour, perceived stress and movement-related outcomes. Viva voce: 6 October 2026; passed with minor corrections; Thesis nominated for Best Thesis Award.

Nazirah Binti Gulam Mohamed | Co-supervisor | PhD (from 2022)
Eccentric exercise, weight loss and physiological and psychological responses among women with obesity.

Wen Feng | Co-supervisor | PhD (from 2022)
Combined exercise methods, dietary habits and health of university students in China.

Zhang Di | Co-supervisor | MSc Research (from 2023)
Sling exercise therapy combined with conventional rehabilitation after meniscus repair.

Li Muzheng | Co-supervisor | MSc Research (2022-2025)
Assessment of balance and falls-prevention knowledge and practices among nursing-home older adults. Graduated in 2025.

CONFERENCE PRESENTATIONS

International and national oral presentations

- 2026 - Hidden Musculoskeletal Vulnerability in Older Women: Integrating Adiposity, Bone Quality, Physical Function, and Vitamin D Bioavailability. Oral presentation, 12th International Conference on Public Health (ICOPH 2026), 5-7 August 2026, Bali Sunset Road Convention Center, Bali, Indonesia. Abstract ID: ICOPH 2026 A 821.
- 2018 - Development of screening test criteria for osteosarcopenia in obese postmenopausal Malaysian women. Nottingham-Adelaide Postgraduate Symposium, University of Nottingham Malaysia.
- 2015 - Associations between accelerometer-determined physical activity and self-reported health-related quality of life in semi-rural older Malay adults. International Conference on Movement, Health and Exercise, Penang.
- 2013 - Preliminary observation on objective measurement of physical activity and sedentary behaviour using tri-axial accelerometers in Malay older adults. International Sports Science Student Conference, University of Malaya.
- 2013 - Are Malaysian elders active? Objective assessment of physical activity in community-dwelling older adults. USM-University of Sydney-University of Queensland Joint Research Seminar, Australia.

Selected poster presentations

- 2022 - Osteosarcopenic obesity and total/bioavailable 25(OH)D in postmenopausal women. International Conference on Frailty and Sarcopenia Research, Boston, USA.
- 2018 - Body fat, serum 25-hydroxyvitamin D and muscle strength in postmenopausal Malaysian women. Nutrition Society of Malaysia Annual Scientific Conference, Kuala Lumpur.
- 2017 - Musculoskeletal health in postmenopausal Malaysian Malay women. Asia Pacific Conference on Clinical Nutrition, Adelaide, Australia.
- 2014 - How active are our Malay elderly population? AMDI Postgraduates Colloquium on Translational Research, Universiti Sains Malaysia.

AWARDS, DISTINCTIONS AND FELLOWSHIP

2026	**Gold Medal (Professional), Virtual Innovation Competition 2026 (VIC2026), for "PhysioVerse 3D: Interactive Exercise Science Learning Platform". Organisers: DIGIT360 and Universiti Teknologi MARA (UiTM).**
2025	**Gold Medal and Special Award First Runner-Up, USM-AUSC International Sports Innovation Challenge (resveratrol microneedle innovation).**
2025	**Silver Award, Micro-credential Category, Academy Fiesta@USM.**
2012-2021	**USM Academic Staff Training Scheme (ASTS) Fellowship.**

INTELLECTUAL PROPERTY AND DIGITAL INNOVATIONS

Registered Malaysian copyright notifications (2026)

CRLY2026E01949 | 9 April 2026 Step-by-Step Visual Protocol for the Fabrication of Resveratrol-Loaded Dissolvable Microneedle Patches. Registered authors: Nurdiana Zainol Abidin et al. Owner: Universiti Sains Malaysia; literary work.

CRLY2026E01727 | 1 April 2026 The Effects of Resveratrol Supplementation on Exercise Performance, Recovery, and Muscle Physiology. Registered author: Nurdiana Zainol Abidin. Owner: Universiti Sains Malaysia; literary work.

Additional registered works

- Cellular Lifespan Part I (film), CRFM2023P02711
- Cellular Lifespan Part II (film), CRM2024P03804
- Cellular Lifespan Part II (film), CRFM2025P07518
- DADU DODECA-FIT (art), CRAR2025P07523

Educational technology and translational innovation

PhysioVerse 3D | Founder and developer (2026)
Interactive web-based 3D exercise science and biomechanics education platform, integrating exercise demonstrations, muscle mapping, joint actions, coaching cues and clinical learning notes. Gold Medal (Professional), VIC2026. Project website

PhysioTwin EDU | Founder and developer (2026)
Interactive teaching application for clinical exercise science and sarcopenia education, case-based assessments and physiological thresholds; in ongoing development and scientific validation.

MSc Clinical Exercise Science Quiz App | Founder and developer (2026)
Web/mobile-accessible postgraduate quiz platform with module-based questions, immediate feedback and student-directed revision. Project website

PRIME-Active | Principal developer (2026)
Active ageing and community physical-activity engagement digital health platform. Prototype stage; co-design and piloting planned.

Microneedle platform invention disclosure | Inventor / research lead (2026)
Solid-state engineered dissolving microneedle platform fabrication. Invention disclosure submitted for USM IP assessment; no patent grant claimed.

TEACHING, CURRICULUM LEADERSHIP AND ACADEMIC MENTORING

Postgraduate teaching TCE504 Applied Exercise Physiology (cellular lifespan, exercise immunology, ageing and neuromuscular/cardiorespiratory responses); TCE515 Clinical Exercise Physiology (neurological, bone and joint conditions); TCE517 Practicum (supervision and coordination); TMR507 Research (internal examination).

Programme and curriculum leadership Head of Programme, MSc Clinical Exercise Science (2023/2024 to 2025/2026); academic coordination, assessments, quality assurance and programme administration. Contributed to the 2026 MSc Clinical Exercise Science curriculum review and related quality-assurance materials.

Micro-credential production Developed nine educational videos, including Foundations to Exercise Physiology and Exercise Physiology for Special Populations.

Academic mentoring Postgraduate advising, research supervision, clinical practicum monitoring and assessment.

UNIVERSITY SERVICE, LEADERSHIP AND COMMUNITY ENGAGEMENT

Project leader, Karnival Warga Emas 2022. Community programme jointly organised with the Ministry of Higher Education Malaysia and Masjid Abdullah Fahim, Penang (12 November 2022).

University committee roles. Academic Quality Committee; Academic Accreditation Committee; Technology Driven Education Committee; Academic Programme Promotion Committee; academic advising.

Conference organisation / chief secretary. International Fasting and Sustainable Health Conference (2010); Malaysia-Australia Research Colloquium on Exercise, Nutrition, Health and Wellness (2011); International Conference on Movement, Health and Exercise (2015).

Research and facilities administration. Laboratory development and record management; community-lifestyle intervention coordination; academic event and international conference organisation.

NEWSPAPER COMMENTARY AND PUBLIC SCIENCE COMMUNICATION

Newspaper opinion articles

The silent health crisis hidden by the scales
Malay Mail, 18 May 2026. Republished in theSun on 30 July 2026. Malay Mail | theSun

Why your muscle matters in the fight against cancer
Malay Mail, 29 May 2026; also published in BusinessToday on 29 May 2026. Malay Mail | BusinessToday

Institutional newsletters and magazine pieces

2025 | Mechanisms of Augmented Reality (AR) and Virtual Reality (VR) on the Human Brain (AMDI Newsletter, 4 June).

2024 | Innovative Healthcare: The Rise of Augmented Reality (AR) and Virtual Reality (VR) (AMDI Newsletter, 2 July).

2024 | Virtual Fitness: Revolutionizing Health and Wellness Through Immersive Exercise (AMDI Newsletter, 10 June).

2022 | The Obesity Paradox: Could Obesity Be Beneficial to You? (AMDI Newsletter, 23 August).

2021 | Total vs. Bioavailable: Determining a Better 25(OH)D Index in Association with Bone Density and Muscle Mass in Postmenopausal Women (AMDI Newsletter, co-authored with SR Mitra).

2011 | Physical Activity in Older Adults (Lifestyle Science Cluster Wellness Bulletin, Issue 4).

2010 | Eat Your Way to Weight Loss (Lifestyle Science Cluster Wellness Bulletin, Issue 3).

SELECTED METHODS, TECHNICAL SKILLS AND PROFESSIONAL DEVELOPMENT

Research and analysis IBM SPSS; ActiGraph/ActiLife accelerometer software; Qualtrics XM; EndNote; Microsoft Word, Excel and PowerPoint.

Teaching and science communication Moodle; Wondershare Filmora; VideoScribe; Audacity; development of micro-credential videos and interactive digital teaching tools.

Selected research and supervisory development USM workshops on research ethics, grants management, intellectual property and postgraduate supervisory practices (2022-2023).

PROFESSIONAL MEMBERSHIPS AND CERTIFICATION

The Physiological Society (UK); Nutrition Society of Malaysia; Malaysia Healthy Ageing Society.

Good Clinical Practice (GCP), Universiti Sains Malaysia, 6-8 May 2010.